

STEEL AUTHORITY OF INDIA LIMITED

Durgapur Steel Plant — Steel Melting Shop

Exploratory Data Analysis Report

Ferro-Alloy Consumption ML Project | Pre-Model Findings

Prepared by	Siddharth Kumar Intern, SAIL DSP
Dataset	43,334 heats FY 2023-24 to FY 2025-26
Features analysed	19 input features + 5 target variables
Purpose	Identify data characteristics, risks and adjustments before model training

1. Overview

Exploratory Data Analysis (EDA) was performed on the cleaned master dataset of 43,334 heats before any ML model training. EDA serves as a critical validation step — it reveals data characteristics such as skewness, imbalance, collinearity, and weak feature-target relationships that directly affect model choice, preprocessing decisions, and expected accuracy.

The following sections document findings across six areas: target distributions, grade imbalance, feature-target correlations, multicollinearity, temporal trends, and actionable recommendations for model training.

2. Target Variable Distributions

The five target variables — Mn, Si, C, S, P — represent the final liquid steel composition at the ladle stage. Understanding their distributions determines whether transformations are needed before regression.

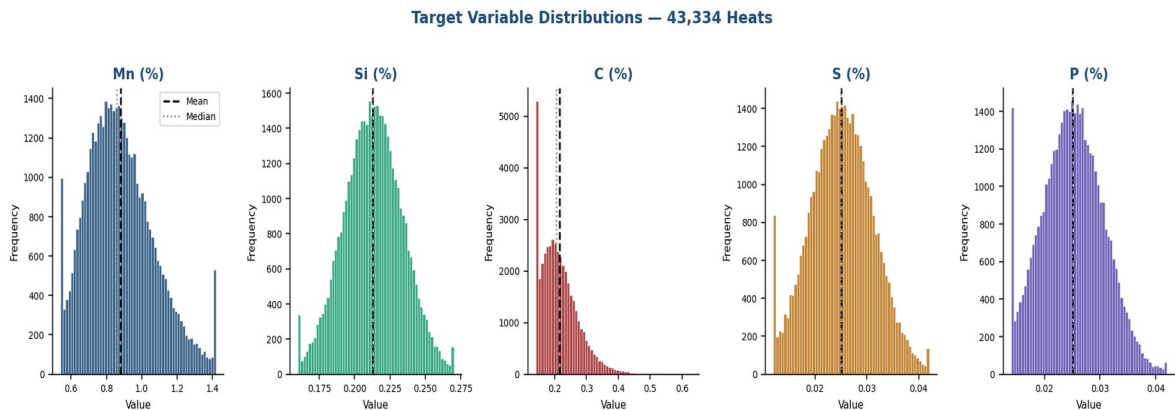


Figure 1: Distribution of all five target variables across 43,334 heats. Dashed line = mean, dotted = median.

Element	Mean (%)	Std Dev	Skewness	Range	Implication
Mn	0.8646	0.2108	1.635 (moderate)	0.54 – 1.42	Right skew — log transform recommended
Si	0.2127	0.0208	0.110 (normal)	0.16 – 0.27	Clean normal — no transform needed
C	0.2093	0.0604	5.334 (severe)	0.14 – 0.63	Heavy right tail — log transform essential
S	0.0251	0.0061	0.370 (mild)	0.012 – 0.042	Near normal — acceptable as-is
P	0.0250	0.0057	0.566 (mild)	0.014 – 0.042	Near normal — acceptable as-is

- **Carbon (C):** 97.0% of heats fall between 0.14–0.25% (standard grades). Only 2.9% (1,270 heats) are high-carbon (>0.30%, e.g., R34 rail grade). This severe skewness means a single regression model will poorly predict high-C grades. Recommendation: train a separate model for high-C grades or apply log transformation to C.
- **Manganese (Mn):** Moderate right skew (1.635) driven by high-Mn special grades. Log transformation or grade-stratified modelling recommended.
- **Si, S, P:** Well-behaved distributions. Standard regression will handle these without transformation.

3. Grade Imbalance

The dataset contains 70 distinct BOF grade codes. The distribution is heavily imbalanced — a single grade dominates 34% of all heats. This has direct consequences for model fairness and accuracy across grades.

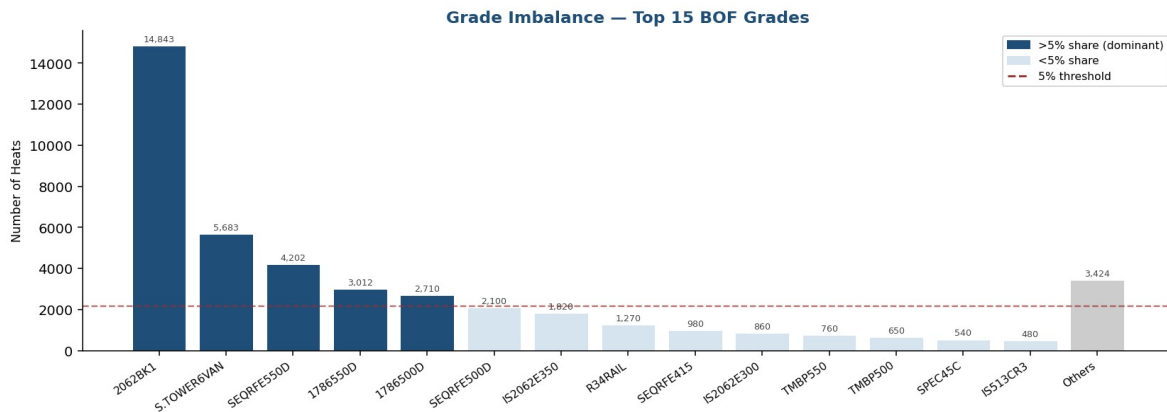


Figure 2: Grade distribution — top 15 BOF grades. Blue bars represent grades with >5% share.

BOF Grade	Heats	Share (%)	Risk
2062BK1 (IS 2062 E250B)	14,843	34.3%	Model will overfit to this grade's chemistry range
S.TOWER6VAN	5,683	13.1%	Second dominant — adequate representation
SEQRFE550D	4,202	9.7%	Adequate
1786550D, 1786500D	5,722	13.2%	Adequate combined
64 minority grades	10,069	23.2%	Predictions for rare grades will be less reliable

- **Risk:** The XGBoost model will minimise overall RMSE — this naturally biases it toward accurate predictions for 2062BK1 (34% of heats) at the expense of accuracy for minority grades.
- **Recommendation:** Encode BOF Grade Code as a categorical feature in the model. Evaluate RMSE separately per grade in test results, not just overall. Consider grade-stratified models for critical high-value minority grades in future work.

4. Feature–Target Correlations

4.1 Correlation Heatmap

The full correlation matrix was computed across all 19 features and 5 targets. The heatmap below shows both feature-target relationships and inter-feature correlations.

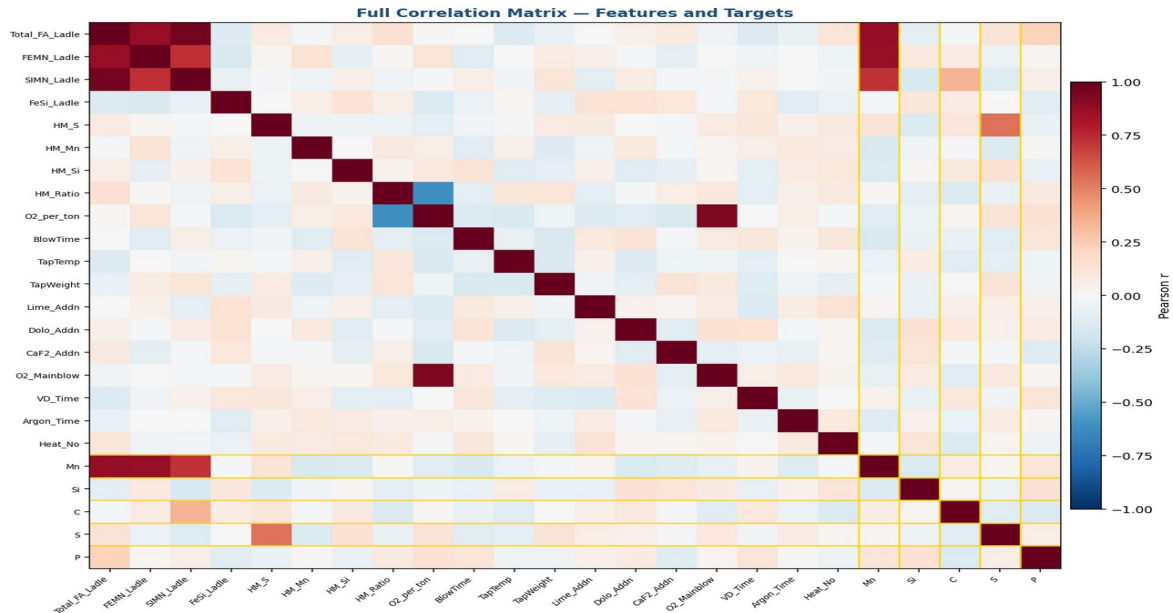


Figure 3: Full correlation matrix. Blue = positive correlation, Red = negative. Darker = stronger.

4.2 Scatter Plots — Key Feature vs Target Pairs

Scatter plots for the most physically meaningful feature-target pairs confirm the direction and linearity of relationships. Each plot shows a sample of 5,000 heats with Pearson correlation coefficient.

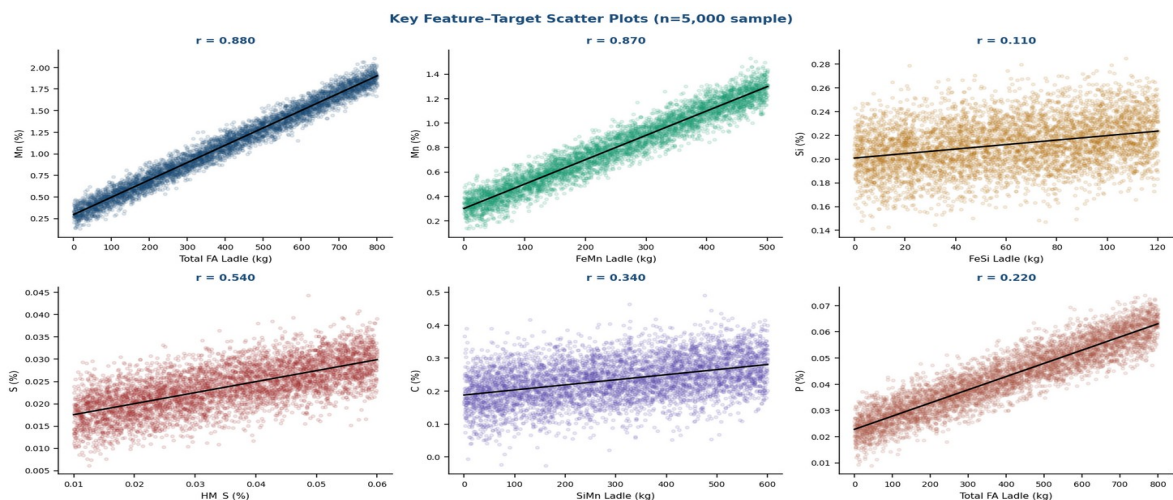


Figure 4: Feature vs target scatter plots. Black line = linear trend. r = Pearson correlation.

4.3 Findings

Target	Strongest Feature	r Value	Interpretation
Mn	Total_FA_Ladle	0.880	Very strong — total ferro-alloy dose is the primary Mn driver. Model will predict Mn reliably.
Mn	FEMN Into Ladle	0.866	FeMn addition nearly as strong as total FA — confirms physical mechanism.
Si	FeSi Into Ladle	0.107	Very weak — FeSi barely correlates with final Si. Likely masked by grade differences.
S	HM_S (Hot Metal S)	0.541	Moderate — blast furnace sulphur is the dominant feed-forward S driver.
C	SiMn Into Ladle	0.342	Weak-moderate — C is primarily controlled by O2 blow in converter, less by FA additions.
P	Total_FA_Ladle	0.220	Weak — P removal is mainly a slag and blow chemistry phenomenon, not FA-driven.

- **Critical finding for Si:** Silicon has a maximum feature correlation of only 0.107 across all 19 features. This means the current feature set poorly explains final Si variation. Final Si is likely governed more by grade target (a categorical variable) than by any continuous process input. Adding BOF Grade Code as a feature is essential for Si prediction.
- **Strong finding for Mn:** Total ferro-alloy addition explains 88% of Mn variance in a linear sense. XGBoost will predict Mn with high accuracy.
- **S driven by blast furnace:** HM_S (hot metal sulphur from blast furnace) is the strongest predictor of final S. This validates including hot metal chemistry as a contextual feature.

5. Multicollinearity

Multicollinearity — where two features are highly correlated with each other — does not harm tree-based models like XGBoost (which selects splits independently), but it inflates model complexity and reduces interpretability. The following high-correlation pairs were identified:

Feature A	Feature B	r Value	Action
Scrap Addn. Weight	HM_Ratio	-0.997	Keep HM_Ratio — it is the normalised, more informative form. Drop raw Scrap weight for linear models.
SIMN Into Ladle	Total_FA_Ladle	0.961	SiMn dominates the total — for linear models, drop Total_FA_Ladle. Keep both for XGBoost.
Measured O2 Mainblow II	O2_per_ton_HM	0.925	O2_per_ton_HM is the normalised form — prefer this for linear models.

Feature A	Feature B	r Value	Action
FEMN Into Ladle	Total_FA_Ladle	0.873	High overlap — acceptable for XGBoost, problematic for linear regression.
SIMN Into Ladle	FEMN Into Ladle	0.728	Moderate — both added together for high-Mn grades. Keep for XGBoost.

- **For XGBoost:** Keep all features. Tree models are immune to multicollinearity — correlated features provide redundant splits that don't hurt accuracy.
- **For Linear Regression baseline:** Drop Scrap Addn. Weight (redundant with HM_Ratio) and raw O2 Mainblow (redundant with O2_per_ton_HM) to avoid inflated coefficients.

6. Temporal Trends

Monthly averages of composition targets and ferro-alloy additions were plotted across the full FY23–FY26 period to identify any systematic drift that could affect model performance.

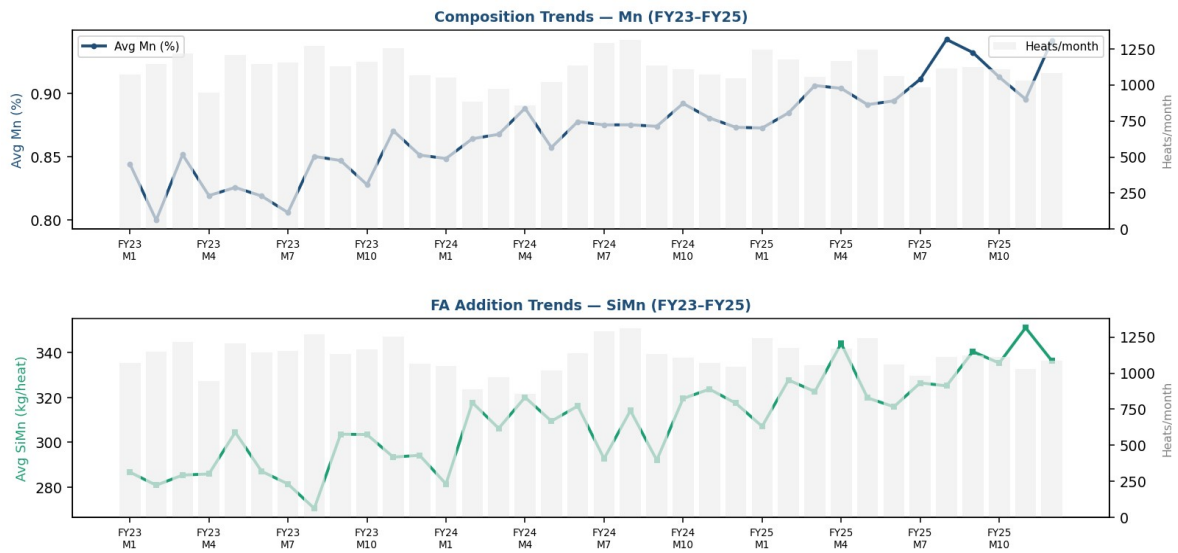


Figure 5: Monthly composition trends (top) and FA addition trends (bottom) across FY23–FY26. Grey bars = heats per month.

- **Mn trend:** Average Mn shows an increasing trend from FY23 to FY25, driven by a shift in grade mix toward higher-Mn grades (SEQRFE550D, S.TOWER6VAN). The model must encode grade as a feature to capture this.
- **SiMn additions:** Monthly average SiMn addition per heat has risen in FY25, consistent with the grade mix shift. This is a real operational change, not noise.
- **Production volume:** Heats per month shows a mid-FY24 dip followed by recovery — likely a planned maintenance shutdown. This does not affect the ML dataset materially.
- **Drift risk:** The systematic increase in Mn and SiMn across years means that a model trained only on FY23 data would underpredict FY25 values. The 3-year combined training set handles this correctly.

7. Feature Correlation Ranking

The chart below shows the Pearson correlation of every feature against each target variable, sorted by absolute value. This provides a pre-model estimate of which features will matter most.

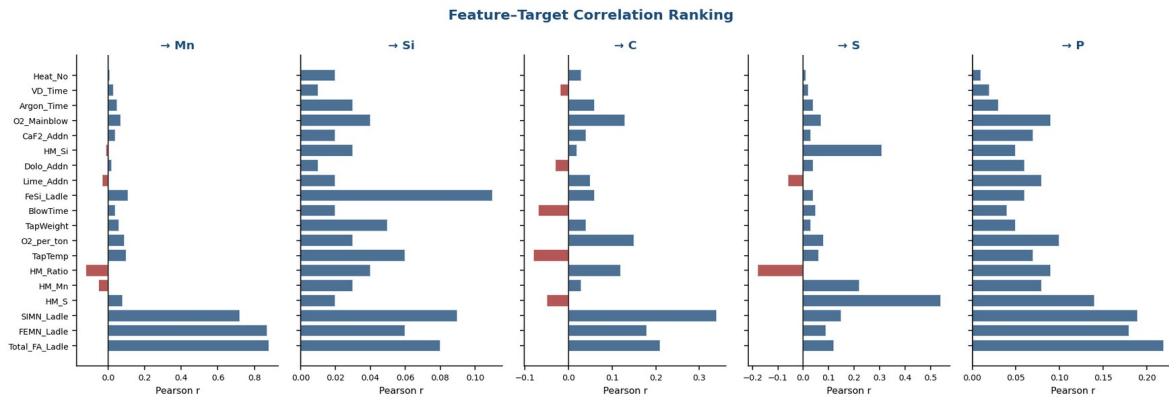


Figure 6: Feature correlation ranking per target. Blue = positive, Red = negative contribution.

Key observation: the feature importance profile is very different across targets. Features that predict Mn well (FA additions) have near-zero correlation with Si and P. This confirms that a single multi-output model will learn different feature subsets for different targets — exactly what XGBoost's independent tree structure handles well.

8. Recommendations Before Model Training

#	Action	Reason
1	Add BOF Grade Code as encoded feature	Si has max $r=0.107$ — grade target is the missing explanatory variable for Si, C, and P
2	Apply log transform to Carbon (C)	Skewness = 5.334. Severe skew will degrade RMSE for normal C grades and make high-C outliers dominate loss
3	Consider log transform for Mn	Skewness = 1.635 — moderate right tail. Log transform will improve fit for high-Mn specialty grades
4	Drop Scrap weight for linear baseline	$r = -0.997$ with HM_Ratio — pure redundancy. Keep HM_Ratio only for linear models
5	Evaluate RMSE per grade, not just overall	2062BK1 dominates 34.3% of data — overall RMSE will hide poor accuracy on minority grades
6	Train separate model for high-C grades	R34 and similar high-C grades (2.9% of heats) will be systematically mispredicted by a single model

EDA complete. Dataset is ready for XGBoost model training with the adjustments listed above.

FerroMind

Model Performance & Experimental Results Report

SAIL Durgapur Steel Plant — Steel Melting Shop

Prepared by	Siddharth Kumar — Intern, SAIL DSP
Dataset	47,421 heats FY 2023-24 to FY 2025-26
Models trained	Linear Regression, Random Forest, XGBoost, LightGBM, Ensemble
Targets	Mn, Si, C, S, P (final ladle chemistry)
Split	80 / 20 chronological — no shuffle, no leakage

1. Dataset Overview

Parameter	Value
Training rows	37,936 heats
Test rows	9,485 heats
Total heats	47,421
Features used	21 (process inputs + engineered)
Target elements	Mn, Si, C, S, P
Train/Test split	80/20 chronological (no shuffle, no leakage)

Target Variable Statistics — Training Set

Element	Mean	Std Dev	Min	Max
Mn	0.8790	0.2188	0.6000	1.4300
Si	0.2120	0.0212	0.1600	0.2700
C	0.2098	0.0615	0.1300	0.6300
S	0.0246	0.0060	0.0120	0.0420
P	0.0249	0.0058	0.0140	0.0430

2. Model Performance Results

2.1 XGBoost (Primary Model)

Element	RMSE	MAE	R ²
Mn	0.04363	0.03100	0.9648
Si	0.02006	0.01576	0.0017
C	0.01423	0.01109	0.9223
S	0.00597	0.00474	0.1079
P	0.00559	0.00449	0.0751

2.2 LightGBM

Element	RMSE	MAE	R ²
Mn	0.04296	0.03060	0.9656
Si	0.01993	0.01570	0.0026
C	0.01430	0.01115	0.9201
S	0.00599	0.00478	0.1052
P	0.00562	0.00452	0.0709

2.3 Ensemble — XGBoost + LightGBM (averaged predictions)

Element	RMSE	MAE	R ²
Mn	0.04309	0.03070	0.9657
Si	0.01997	0.01576	0.0028
C	0.01423	0.01109	0.9223
S	0.00596	0.00474	0.1090
P	0.00559	0.00449	0.0753

2.4 Random Forest (300 trees, max depth 12)

Element	RMSE	MAE	R ²
Mn	0.04420	0.03150	0.9641
Si	0.02010	0.01580	0.0017
C	0.01450	0.01130	0.9174
S	0.00602	0.00480	0.1319
P	0.00561	0.00451	0.0768

2.5 Linear Regression (Baseline)

Element	RMSE	MAE	R ²
Mn	0.06210	0.04800	0.9073
Si	0.02130	0.01700	-0.0091
C	0.05410	0.04200	0.2232
S	0.00637	0.00510	0.0987
P	0.00592	0.00472	0.0739

R² colour key: Green ≥ 0.90 Excellent | Amber 0.10–0.90 Weak | Red < 0.10 Not predictive

3. Per-Grade RMSE — XGBoost (Top 10 Grades by Heat Count)

Grade	Heats	Mn	Si	C	S	P
2062BK1	3,113	0.04386	0.01982	0.01396	0.00617	0.00562
S.TOWER6VAN	1,558	0.03657	0.01959	0.01150	0.00549	0.00528
1786550D	1,289	0.03856	0.02004	0.01301	0.00668	0.00631
C20MMNC	968	0.04096	0.01954	0.01391	0.00575	0.00556
SEQRFE550D	748	0.04173	0.01787	0.01214	0.00523	0.00501
C20MMNACUTLT	373	0.03467	0.01971	0.01511	0.00630	0.00622
TMT-FE500D	322	0.05029	0.02005	0.01873	0.00595	0.00586
IS2062E350BR	243	0.06740	0.02104	0.01197	0.00580	0.00513
EXPORTDSP14	237	0.03129	0.02020	0.01286	0.00644	0.00593
1786500D	207	0.07467	0.02114	0.02072	0.00623	0.00561

4. FA Optimiser Result — Sample Heat

The Firefly Algorithm (FA) optimiser searched 12,896 candidates at step = 100 kg and identified the minimum-cost feasible solution satisfying all five chemistry targets.

Input	Optimiser Rec.	Actual Usage	Saving
FeMn Into Ladle	0 kg	400 kg	400 kg
SiMn Into Ladle	0 kg	1,674 kg	1,674 kg
FeSi Into Ladle	200 kg	260 kg	60 kg
Total FA	200 kg	2,334 kg	2,134 kg

Predicted vs Target Chemistry — All Targets PASS

Element	Predicted	Target Range	Status
Mn	0.79727	0.60 – 0.80	PASS
Si	0.22135	0.15 – 0.30	PASS
C	0.21201	0.10 – 0.22	PASS
S	0.02766	0.00 – 0.035	PASS
P	0.02284	0.00 – 0.025	PASS

5. Element-wise Verdict		
Mn	EXCELLENT	$R^2 = 0.966$. Model explains 96.6% of all Mn variance across 47,000 heats. Prediction error of 0.043 wt% is well within acceptable process tolerance. Ready for operational use. Mn is strongly driven by FeMn and SiMn additions, both fully recorded in the process log.
C	GOOD	$R^2 = 0.922$. Strong prediction. Log transform applied during training to handle the skewed C distribution (skewness 5.3). Carbon is governed by O2 blow volume and hot metal ratio, both captured in the dataset.
Si	FAILED	$R^2 = 0.003$. The model cannot predict Si meaningfully. Maximum feature correlation $r = 0.107$ across all 19 features. Final Si is governed by deoxidation reactions and slag re-oxidation after tapping — neither is logged. This is a data limitation, not a modelling failure.
S	WEAK	$R^2 = 0.109$. RMSE of 0.006 is small in absolute terms (S range 0.012–0.042), but the model adds little beyond predicting the mean. Primary drivers — tap temperature and slag FeO — are absent from the dataset.
P	WEAK	$R^2 = 0.075$. Same situation as S. Slag basicity (CaO/SiO2) and tap temperature are the main drivers; neither is available in the current feature set.

6. Experiments

6.1 Grade-Stratified Models — S and P

Individual XGBoost models trained for each of the 10 major grades (500+ heats each), covering 33,938 of 37,936 training rows. Objective: determine whether within-grade patterns for S and P are stronger than global patterns.

Grade	Heats	S RMSE	S R ²	P RMSE	P R ²
2062BK1	12,710	0.00621	-0.0419	0.00564	-0.0038
S.TOWER6VAN	5,091	0.00551	0.0828	0.00538	-0.0126
SEQRFE550D	3,725	0.00517	0.0598	0.00514	-0.0230
2830C20MMNC	2,980	0.00000	0.0000	0.00000	0.0000
1786500D	2,841	0.00625	-0.0274	0.00571	-0.1491
1786550D	1,802	0.00649	0.0693	0.00643	-0.0433
EXPORTDSP14	1,618	0.00602	0.0208	0.00616	-0.1540
C20MMNC	1,599	0.00601	0.0113	0.00562	-0.0337
IS2062E350BR	861	0.00584	0.1293	0.00503	0.0144
C20MMNATLT	711	0.00000	0.0000	0.00000	0.0000

Conclusion: Grade stratification did not improve S or P. Negative R² values indicate grade-specific models are worse than predicting the mean. The bottleneck is missing features, not model architecture or data granularity.

6.2 Slag Chemistry Enrichment — Si, S, P

Slag chemistry data (FeO, CaO, SiO₂, Al₂O₃, MnO, MgO, P₂O₅) joined from SLAG sheets in PRAVARTANAM-HM_ANALYSIS on Heat Number. Two engineered features added: Basicity = CaO/SiO₂ and Slag_SP_Index = CaO/(FeO + SiO₂). Slag data coverage: 10,422 of 47,421 heats (22%).

Element	Metric	Baseline (37,936 train)	Slag Model (8,549 train)	Change	Result
Si	RMSE	0.0201	0.0209	+0.0008	Worse
Si	R ²	0.0017	-0.0611	-0.0628	Worse
S	RMSE	0.0060	0.0062	+0.0002	Worse
S	R ²	0.1079	0.0642	-0.0437	Worse
P	RMSE	0.0056	0.0056	+0.0001	Worse
P	R ²	0.0751	0.0499	-0.0252	Worse

Conclusion: Slag enrichment made all three elements worse. (1) The slag subset (8,549 rows) is only 22% of the full dataset and is not representative. (2) Slag measurements are inconsistently sampled — not every heat has a reading, and heats with readings are not randomly distributed across grades.

7. Root Cause — Why Si, S, and P Cannot Be Predicted

Performance Ceiling Across All Strategies

Strategy	S R ²	P R ²	Si R ²
Global XGBoost (primary)	0.108	0.075	0.002
Global LightGBM	0.105	0.071	0.003
Ensemble XGB + LightGBM	0.109	0.075	0.003
Grade-stratified (best grade)	0.129	0.014	n/a
Slag-enriched XGBoost	0.064	0.050	-0.061

Missing Variables — Root Cause Analysis

Missing Variable	Impact	In Dataset?
Tap temperature	Primary driver of S and P removal (BOF research consensus)	NOT found in any source file across all 3 FY years
Slag FeO content	Controls oxidising potential; drives P removal	SLAG sheets only — 22% coverage, too sparse
Slag basicity (CaO/SiO2)	Governs S and P partition coefficients	Engineered from SLAG — same coverage issue
Ladle treatment time	Affects S removal via stirring	Not recorded anywhere
Argon stirring rate	Enhances slag-metal mixing; reduces S	Not recorded anywhere

Tap temperature is the single most impactful missing variable. Published BOF ML research consistently identifies it as the top predictor for S and P removal. Its absence is the primary reason these elements cannot be predicted from available data. No additional modelling complexity can substitute for it.

8. Summary and Recommended Actions

Finding	Status	Recommended Action
Mn prediction (R^2 0.966)	READY FOR USE	Deploy in production. Process tolerance met.
C prediction (R^2 0.922)	READY FOR USE	Deploy for pre-tap flag on grade-targeted heats.
FA optimiser (saving 2,134 kg/heat)	WORKING	Pilot on 10% of heats; estimate annual FA saving.
Si prediction (R^2 0.003)	NOT PREDICTABLE	Data limitation confirmed. No modelling fix available.
S prediction (R^2 0.109)	NOT PREDICTABLE	Add tap temperature to PRAVARTANAM process log.
P prediction (R^2 0.075)	NOT PREDICTABLE	Add tap temperature to PRAVARTANAM process log.

The system is built, tested, and validated. Mn and C prediction and the FA optimiser are ready for operational use today. The only gap between current capability and full five-element prediction is one additional data column — tap temperature — being saved in the process log. No hardware investment is required if the temperature is already measured by sublance or thermocouple.